

$\Rightarrow$

0

WA

0

DE-13/05/26

$$e^{\gamma t} (m\gamma^2 + b\gamma + k) = 0 \quad | \quad \underline{\underline{x(t) = e^{\gamma t}}}$$

$\therefore e^{\gamma t} \neq 0$ as then $x(t) = 0$ always.

then $m\gamma^2 + b\gamma + k = 0$

$\Rightarrow \gamma = \dots\dots\dots$

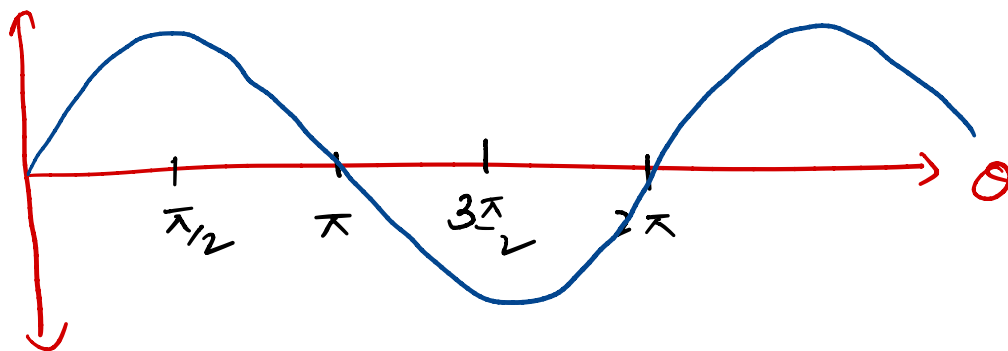
Time Period and frequency

$$x(t) = A \sin(\omega t + \alpha)$$

$$= A \sin \theta \quad ; \quad \theta \text{ is } t \text{ dependent}$$

Time period $\rightarrow T \rightarrow$ In T time, one full oscillation will be completed.

$$\therefore x(t+T) = x(t)$$



$\sin \theta$ — 24 Period = 2π 5121, $\sin(\theta + 2\pi) = \sin \theta$

$\therefore x(t+T) = x(t)$ 212 0 512 252,

Sin-function ki 2π -cycle complete hoga,

$$x(t+T) = A \sin [\omega(t+T) + \alpha]$$

$$\equiv A \sin (\omega t + \alpha + 2\pi)$$

$$= A \sin (\omega t + \alpha)$$

$$\therefore \omega(t+T) + \alpha = \omega t + \alpha + 2\pi$$

$$\Rightarrow \omega T = 2\pi$$

$$\Rightarrow T = \frac{2\pi}{\omega}$$

$\therefore \frac{2\pi}{\omega}$ time $\rightarrow$ 1 full cycle of oscillation

Complete 2π ,

Frequency :- no. of complete oscillation
in unit time.

$$\therefore \text{frequency, } n = \frac{1}{\text{Time Period}} = \frac{1}{T} = \frac{1}{\frac{2\pi}{\omega}}$$

$$n = \frac{\omega}{2\pi}$$

So, ω is basically an angular velocity.

If a particle is moving in a circular motion.

Then, one complete cycle means $\Rightarrow 2\pi$ rad.
angular displacement.

Now, if ω is its angular velocity, that means it goes ω rad. per sec.

Again, 2π rad represents 1 complete cycle.

$\therefore$ no. of cycles completed in unit time.

$$= \frac{\omega}{2\pi}$$

Also, for SHM, frequency = $\frac{\omega}{2\pi}$.

So, here also, ω is an angular velocity analogue.
Unit of $\omega \Rightarrow \text{rad} \cdot \text{s}^{-1}$

Phase and Phase Difference:

$$x(t) = A \sin(\omega t + \alpha)$$

$$= A \sin \theta$$

So, θ here represents a which "Phase" of the motion the particle is.

$$\frac{x(t)}{A} = \sin \theta$$

$\theta(t) \rightarrow$ Phase of oscillation.

The initial phase $\Rightarrow \theta(t=0) = \omega \cdot 0 + \alpha = \alpha$

- is called the epoch.

Phase Difference:

2 SHO $\Rightarrow$

$$x_1(t) = A_1 \sin \theta_1(t)$$

$$x_2(t) = A_2 \sin \theta_2(t)$$

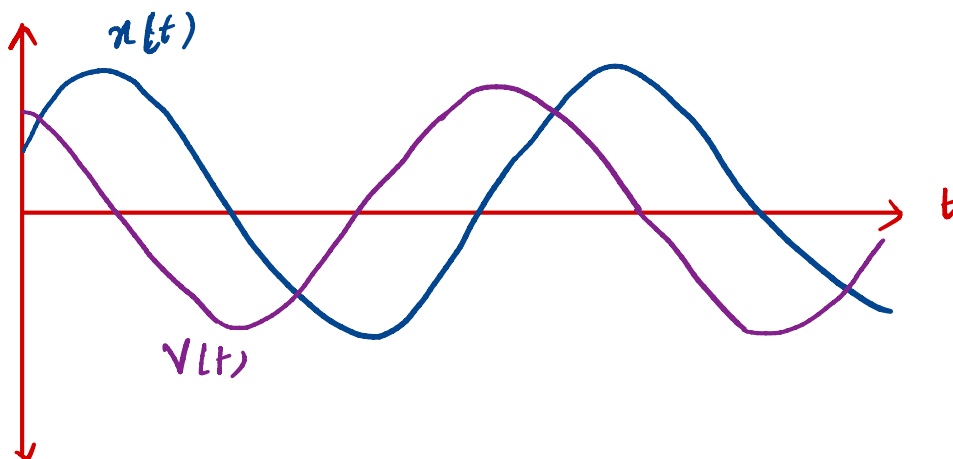
Phase Difference at time $t \Rightarrow \theta_1(t) - \theta_2(t)$

" " at initial time $\Rightarrow \theta_1(0) - \theta_2(0)$

Plot of $x(t)$ and $v(t)$:

$$x(t) = A \sin(\omega t + \alpha)$$

$$; v(t) = A\omega \cos(\omega t + \alpha)$$



* There is a $\frac{\pi}{2}$ - phase difference between x and v .

* When $x = \pm A \Rightarrow v = 0$
(Max.)

* When $x = 0 \Rightarrow v = \pm A\omega$ (Max.)

Numerical Problems:

Q11 A particle going through SHO has displacement,

$$x(t) = 3 \sin 60\pi t$$

What is its amplitude, Time period and acceleration at maximum displacement?

Ans.

$$x(t) = 3 \sin (60\pi t) \equiv A \sin(\omega t + \alpha)$$

Amplitude, $A = 3$

Angular freq., $\omega = 60\pi$

$\therefore$ Time period, $T = \frac{2\pi}{\omega}$

$$= \frac{2\pi}{60\pi} = \frac{1}{30}$$

acceleration, $a = -\omega^2 x$

acceleration at Max. displacement

$$a = -\omega^2 \cdot 3 \quad (\text{max, } x_{\text{max}} = 3)$$

$$= -(60\pi)^2 \cdot 3$$

$$= -10800\pi^2$$

Q12 If, $x = 4(\sin \pi t + \cos \pi t)$ cm, then what is the amplitude?

Ans. $x = 4(\sin \pi t + \cos \pi t)$ cm.

$$= 4\sqrt{2} \left(\frac{1}{\sqrt{2}} \sin \pi t + \frac{1}{\sqrt{2}} \cos \pi t \right) \text{ cm.}$$

$$= 4\sqrt{2} \left(\cos \frac{\pi}{4} \sin \pi t + \sin \frac{\pi}{4} \cos \pi t \right)$$

$$= 4\sqrt{2} \sin \left(\pi t + \frac{\pi}{4} \right) \text{ cm.}$$

$$\therefore \text{Amplitude} = 4\sqrt{2} \text{ cm.}$$

Q3] The equation of displacement of an SHO is,

$$x(t) = 10 \sin \left(\frac{\pi}{3} t - \frac{\pi}{12} \right) \text{ cm.}$$

calculate its $\rightarrow$ i) Amplitude

ii) Time period

iii) Highest velocity

iv) " Acceleration

v) Initial phase

vi) vel. at 1 sec.

Ans. $x = 10 \sin \left(\frac{\pi}{3} t - \frac{\pi}{12} \right) \text{ cm} \equiv A \sin(\omega t + \alpha)$

$$\therefore A = 10 \text{ cm}$$

$$\omega = \frac{\pi}{3} \text{ rad. s}^{-1}$$

$$\alpha = -\frac{\pi}{12}$$

i) Amplitude, $A = 10 \text{ cm}$

ii) Time Period, $T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{\pi}{3}} \text{ s}$
 $= 6 \text{ s}$

iii) Velocity, $V = \pm \omega \sqrt{A^2 - x^2}$

$$\therefore V_{\text{max}} = \pm \omega \sqrt{A^2 - 0} = \pm A\omega = \pm \left(10 \cdot \frac{\pi}{3} \right) \text{ cm.s}^{-1}$$

$$= 1 \frac{10\pi}{3} \text{ cm.s}^{-1}$$

iv) Acceleration, $a = -\omega^2 x$

$$a_{\max} = -\omega^2 x_{\max}$$

$$= -\left(\frac{\pi}{3}\right)^2 \cdot 10 \text{ cm.s}^{-2}$$

$$= -\frac{10\pi^2}{9} \text{ cm.s}^{-2}$$

v) Initial phase, $\alpha = -\frac{\pi}{12}$

vi)
$$v = A\omega \cos\left(\frac{\pi}{3}t - \frac{\pi}{12}\right)$$

$$= 10\frac{\pi}{3} \cos\left(\frac{\pi}{3}t - \frac{\pi}{12}\right)$$

$$v(t=1) = \frac{10\pi}{3} \cos\left(\frac{\pi}{3} - \frac{\pi}{12}\right) = \dots$$

Q41 A particle undergoing SHO completes 60 oscillations in 2s. Initially it was 5 cm away from equilibrium with zero velocity. What is its eqⁿ of displacement.

Ans. Let $x(t) = A \sin(\omega t + \alpha)$

Frequency, $n = \frac{60}{2} \text{ s}^{-1} = 30 \text{ s}^{-1}$

$$\therefore \omega = 2\pi n = 60\pi \text{ rad.s}^{-1}$$

$$x(t) = A \sin(60\pi t + \alpha)$$

$$\therefore v(t) = A \cdot (60\pi) \cos(60\pi t + \alpha)$$

$$\text{at } t=0 \Rightarrow x(0) = A \sin(\alpha) = 5$$

$$v(0) = 60\pi \cdot A \cdot \cos(\alpha) = 0$$

$$\therefore \cos \alpha = 0$$

$$\Rightarrow \alpha = \frac{\pi}{2}$$

$$\text{Again, } A \sin(\alpha) = 5$$

$$A \sin\left(\frac{\pi}{2}\right) = 5$$

$$\Rightarrow A = 5$$

$$\therefore x(t) = 5 \sin(60\pi t + \frac{\pi}{2})$$

Q51 The frequency of oscillation of a wire is 200 Hz. A point on the wire has velocity = 4.35 m/s⁻¹ when it reaches half its amplitude. Calculate the acceleration of that point at that time.

$$\text{Ans. acceleration, } a = -\omega^2 x$$

$$\therefore \text{At half its amplitude, } a_{\frac{x}{2}} = -\omega^2 \cdot \frac{A}{2} \quad (A \rightarrow \text{amplitude})$$

$$\text{Now, frequency, } n = 200 \text{ s}^{-1}$$

$$\therefore \omega = 2\pi n = 400\pi \text{ rad.s}^{-1}$$

$$\text{Now, vel. , } v = \pm \omega \sqrt{A^2 - x^2}$$

$$\therefore \text{vel. at half amplitude, } v = \pm \omega \sqrt{A^2 - \left(\frac{A}{2}\right)^2}$$

$$\text{As, vel. is +ve, } +\omega \sqrt{A^2 - \frac{A^2}{4}} = 4.35$$

$$\Rightarrow 400\pi \cdot \frac{\sqrt{3}}{2} \cdot A = 4.35$$

$$\Rightarrow A = \frac{4.35}{200\pi\sqrt{3}}$$

$\therefore$ acceleration at half amplitude,

$$a_{1/2} = -\omega^2 A$$

$$= - (400\pi)^2 \cdot \frac{4.35}{200\pi\sqrt{3}}$$

$$= - \frac{800\pi}{\sqrt{3}} \times 4.35 \text{ m.s}^{-2}$$

Q1 A particle undergoing SHO has position = 6 cm, 10 cm and 6 cm at 3 consecutive seconds in time. Determine its frequency.

Ans. $x(t) = A \sin(\omega t + \alpha)$

Let's take 3 time points at $t, t+1, t+2$ $\alpha \xrightarrow{+ \omega t} \omega t + \alpha$

$$\therefore 6 \text{ cm} = A \sin(\omega t + \alpha)$$

$$10 \text{ cm} = A \sin(\omega(t+1) + \alpha) = A \sin(\omega t + \omega + \alpha)$$

$$6 \text{ cm} = A \sin(\omega(t+2) + \alpha) = A \sin(\omega t + 2\omega + \alpha)$$

$$A \sin(\omega t + \omega + \alpha) = A [\sin(\omega t + \alpha) \cdot \cos \omega + \cos(\omega t + \alpha) \cdot \sin \omega]$$

Now, $\sin(\omega t + \alpha) = \frac{6}{A}$

$$\sin(\omega t + \alpha) = \frac{6}{A}$$

$$\cos(\omega t + \alpha) = \sqrt{1 - \left(\frac{6}{A}\right)^2} = \sqrt{1 - \frac{36}{A^2}} = \frac{1}{A} \sqrt{A^2 - 36}$$

$$\begin{aligned}
 \therefore A \sin(\omega t + \omega + \alpha) &= A \left[\frac{6}{A} \cdot \cos \omega + \frac{1}{A} \sqrt{A^2 - 36} \cdot \sin \omega \right] \\
 &= [6 \cos \omega + \sqrt{A^2 - 36} \sin \omega] \\
 &= 10
 \end{aligned}$$

Again $A \sin(\omega t + 2\omega + \alpha)$

$$= A [\sin(\omega t + \omega + \alpha) \cos \omega + \cos(\omega t + \omega + \alpha) \sin \omega]$$

$$= A \left[\frac{10}{A} \cos \omega + \sqrt{1 - \left(\frac{10}{A}\right)^2} \sin \omega \right]$$

$$= [10 \cos \omega + \sqrt{A^2 - 100} \sin \omega]$$

$$= 6$$

$A \sin(\omega t + 2\omega + \alpha)$

$$= A [\sin(\omega t + \alpha) \cos(2\omega) + \cos(\omega t + \alpha) \sin(2\omega)]$$

$$= A \left[\frac{6}{A} \cos(2\omega) + \sqrt{1 - \left(\frac{6}{A}\right)^2} \sin 2\omega \right]$$

$$= [6 \cos(2\omega) + \sqrt{A^2 - 36} \sin 2\omega]$$

$$= 6$$

$$6 \cos \omega + \sqrt{A^2 - 36} \sin \omega = 10 \rightarrow \textcircled{1}$$

$$10 \cos \omega + \sqrt{A^2 - 100} \sin \omega = 6 \rightarrow \textcircled{2}$$

$$6 \cos(2\omega) + \sqrt{A^2 - 36} \sin 2\omega = 10 \rightarrow \textcircled{3}$$

$$\textcircled{1} \times 2 \cos \omega$$

$$12 \cos^2 \omega + \sqrt{A^2 - 36} 2 \sin \omega \cos \omega = 10 \times 2 \cos \omega$$

$$\Rightarrow 12 \cos^2 \omega + \sqrt{A^2 - 36} \sin 2\omega = 20 \cos \omega$$

$$\text{From } \textcircled{3}$$

$$\sqrt{A^2 - 36} \sin \omega = 10 - 6 \cos(2\omega)$$

$$\text{So, } 12 \cos^2 \omega + 10 - 6 \cos(2\omega) = 20 \cos \omega$$

$$\Rightarrow 12 \cos^2 \omega + 10 - 6[2\cos^2 \omega - 1] = 20 \cos \omega$$

$$\Rightarrow 12 \cos^2 \omega + 10 - 12 \cos^2 \omega + 6 = 20 \cos \omega$$

$$\Rightarrow 16 = 20 \cos \omega$$

$$\Rightarrow \cos \omega = \frac{16}{20} = \frac{4}{5}$$

$$\Rightarrow \omega = \cos^{-1}\left(\frac{4}{5}\right) = \dots$$

$$\therefore \text{Frequency, } n = \frac{\omega}{2\pi} = \dots$$

An important trick we can use:

$$x(t) = A \sin(\omega t + \alpha)$$

If we shift our time origin to t' such that,

$$t' = t + \beta$$

$$x(t) = A \sin(\omega t + \alpha)$$

$$= A \sin(\omega(t' - \beta) + \alpha)$$

$$= A \sin(\omega t' + \alpha - \omega\beta)$$

$$= A \sin(\omega t' + \alpha')$$

The only thing that changes is the initial phase.

$$\alpha \rightarrow \alpha - \omega\beta$$

So, for that last numerical problem, although the exact 3 time points were not given, but we can choose them to be, $t = 0, 1, 2$.

By doing that, we know we will mis/calculate the initial phase α . But we are not needed to calculate α for that problem.

$$\text{So, } x(t) = A \sin(\omega t + \alpha)$$

$$\therefore 6 = A \sin(\alpha)$$

$$10 = A \sin(\omega + \alpha)$$

$$6^2 = A \sin(2\omega + \alpha)$$



$$A \sin(\omega t + \alpha)$$

Mechanical Energy of an SHO :-

Kinetic Energy

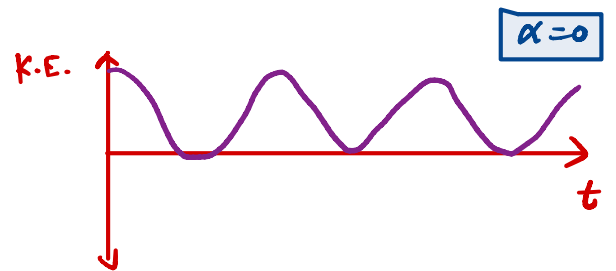
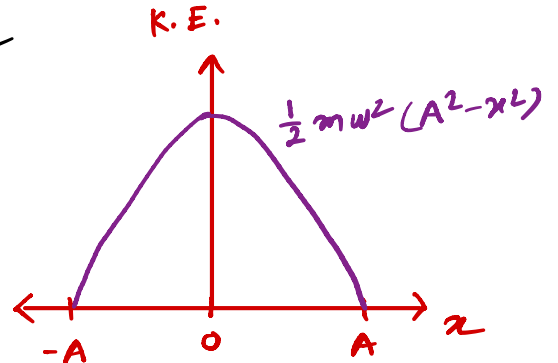
$$\begin{aligned} \text{K.E.} &= \frac{1}{2} m v^2 \\ &= \frac{1}{2} m (\pm \omega \sqrt{A^2 - x^2})^2 \end{aligned}$$

$$\text{K.E.} = \frac{1}{2} m \omega^2 (A^2 - x^2)$$

$$= \frac{1}{2} m \omega^2 [A^2 - A^2 \sin^2(\omega t + \alpha)]$$

$$\text{K.E.} = \frac{1}{2} m \omega^2 A^2 [1 - \sin^2(\omega t + \alpha)]$$

$$y = x^2$$

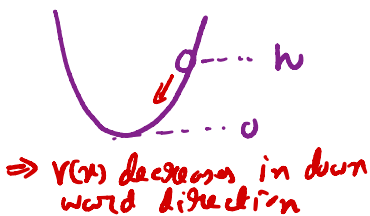


Potential Energy :-

Note :- If at a location x , force is $F(x)$ and Potential Energy is $V(x)$, then,

$$F(x) = - \frac{dV(x)}{dx}$$

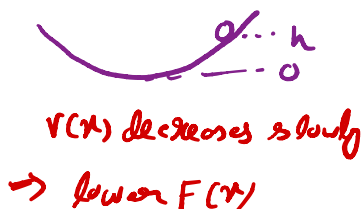
Physical Meaning :- Every body has a tendency to decrease its Potential Energy.



→ The driving force to do that is the force considered here.

→ Now, $\frac{dV}{dx}$ represent increase of P.E. V w.r.t. x .

→ So, Force should always work on the opposite direction i.e. in the direction where P.E., V decreases.



→ higher the rate of decrease of P.E. $V(x)$ along a particular direction, higher will be the force along that direction.

So, now for SHO

$$F(x) = -kx = -m\omega^2 x$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$k = m\omega^2$$

Now, $F(x) = -\frac{dV(x)}{dx}$

$$\Rightarrow V(x) = -\int F(x) dx$$

$$= -\int (-m\omega^2 x) dx$$

$$= m\omega^2 \int x dx$$

$$V(x) = \frac{1}{2} m\omega^2 x^2 + c$$

Now, Potential energy always has an arbitrary reference point, which we can choose according to our convenience.

Let's set, our equilibrium point to be the P.E. = 0 point.

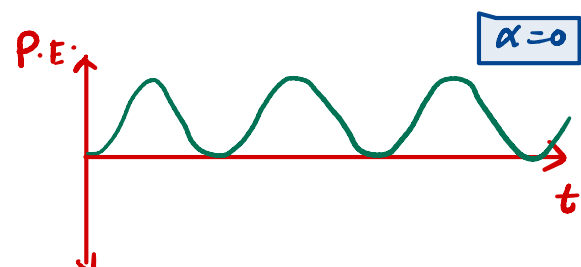
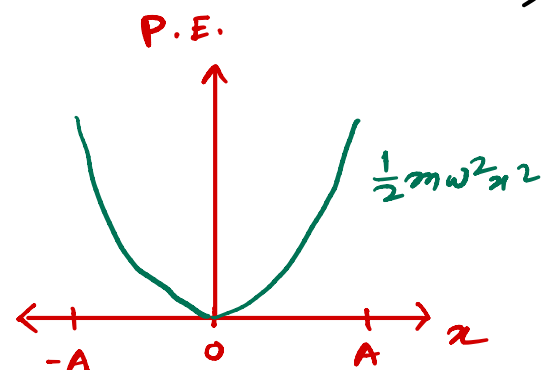
Now at eq^m, $x = 0 \Rightarrow V(x) = 0$ (we have chosen)

$$\therefore 0 = \frac{1}{2} m\omega^2 \cdot 0^2 + c$$

$$\Rightarrow c = 0$$

$$\therefore V(x) = \frac{1}{2} m\omega^2 x^2$$

$$V(x) = \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \alpha)$$

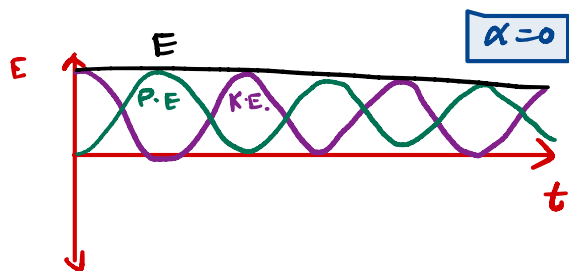
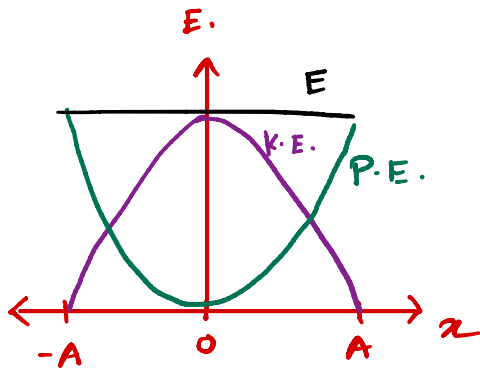


So Total Mechanical Energy,

$$E = K.E. + P.E.$$

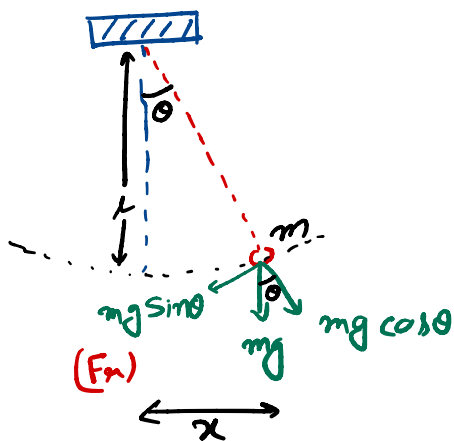
$$= \frac{1}{2} m \omega^2 (A^2 - x^2) + \frac{1}{2} m \omega^2 x^2$$

$$E = \frac{1}{2} m \omega^2 A^2 \Rightarrow \text{Time - independent}$$



Example 5 Hooke's

1. Simple Pendulum:



$$F_n = -mg \sin \theta$$

$$= -mg \theta \quad (\text{when } \theta \text{ is small})$$

$$= -mg \frac{x}{l}$$

$$F_n = -\frac{mg}{l} \cdot x$$

$$\therefore K = \frac{mg}{l}$$

(or), $F = -Kx$
From Hooke's law

$$\therefore \omega = \sqrt{\frac{K}{m}}$$

$$= \sqrt{\frac{g}{l}}$$

$$\therefore T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g}} \quad (g, l \rightarrow \text{constant})$$

* So, the time to complete 1 full cycle only depends on l.

